Better Breathing Better Living ♡

MEDICOS MINDS

Move Better Breathe Better Live Better ♡

PHYSIOTHERAPY IN CARDIOPULMONARY CONDITIONS

EVIDENCE-BASED CARE FOR A HEALTHIER HEART & LUNGS



KEY TOPICS COVERED

- Cardiopulmonary Anatomy & Physiology
- Assessment & Evaluation
- Physiotherapy in Cardiac Conditions
- Physiotherapy in Pulmonary Conditions
- Airway Clearance Techniques
- Respiratory & Cardiac Rehabilitation
- Exercise Prescription & Progression
- Patient Education & Self-Management
- Evidence-Based Practice

Better Function Better Life ♡

LEARNING OUTCOMES

- ✓ Understand cardiopulmonary anatomy & pathophysiology
- ✓ Perform appropriate assessment and clinical reasoning
- ✓ Apply evidence-based physiotherapy interventions
- ✓ Develop individualized rehabilitation plans
- ✓ Improve patient outcomes and quality of life

- CARDIAC CARE
- RESPIRATORY CARE
- REHABILITATION
- AIRWAY CLEARANCE
- EXERCISE THERAPY
- PATIENT EDUCATION

FOR ALL MEDICAL & ALLIED HEALTH STUDENTS

MBBS | BHMS | BSc. NURSING | BPT

Every Breath Counts Every Movement Matters ♥

MEDICOS_MINDS | Learn | Practice | Heal

PHYSIOTHERAPY IN CARDIOPULMONARY CONDITIONS
MEDICOS MINDS

PT in cardio Pulmonary conditions

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OFFICIAL DIGITAL STUDY GUIDE VERIFICATION

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STUDY GUIDE DETAILS & CONTENTS:

Comprehensive high-yield notes designed for Bachelor of Physiotherapy (BPT), MBBS, BDS, and allied healthcare students. Covers fundamental anatomy, clinical assessments, pathological conditions, evidence-based physical therapy rehabilitation protocols, and exam-oriented high-yield review.

KEY HIGHLIGHTS:

- Exam-Focused: Concise points curated from leading medical textbooks and clinical references.
- Visual Diagrams: High-yield diagrams and tables for quick revision.
- Mobile & Tablet Optimized: Readable on phones, tablets, iPads, and laptops.
- Official Medicos Minds Release: Verified by Dr. MD Kamran Alam.



Breathe Better Live Stronger

PHYSIOTHERAPY IN CARDIOPULMONARY CONDITIONS



Healthy Heart Healthy Lungs

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1. Review of Anatomy and Physiology of the Cardio Respiratory System.
2. Patient assessment – Bedside assessment of patient.
3. Investigations and tests – Exercise tolerance test.
4. ECG.
5. ABG.
6. Haematological and biochemical tests.
7. Physiotherapy techniques to increase lung volume – controlled mobilization.
8. Positioning.
9. Breathing exercises.
10. Incentive spirometry.
11. To decrease work of breathing – positioning, breathing re-education, breathing controll techniques.
12. Mechanical aids – IPPB, CPAP, BiPAP.
13. Humidification and nebulisation.
14. Mobilization and breathing exercises.
15. Postural drainage.
16. Manual techniques – percussion, vibration and shaking.
17. Autogenic drainage.
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19. Physiotherapy management in breathlessness.
20. Thrombosis, Embolism, Arteriosclerosis.
21. Thrombophlebitis, Gangrene, Congestive cardiac failure.
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29. Graduated exercise programme and posture correction.
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36. Pulmonary rehabilitation and cardiac rehabilitation.



For All Medical Students

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Topic 4

ECG
(Electrocardiogram)

Heart's Electrical Story

1. What is ECG?

- ECG is a non-invasive test that records the electrical activity of the heart through electrodes placed on the skin.
- It helps to assess heart rate, rhythm, conduction, ischemia/infarction, arrhythmias and some chamber enlargement.

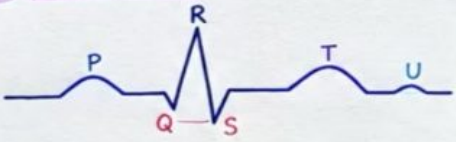
2. Principle

- Heart produces electrical impulses → depolarization & repolarization.
- These changes spread through the heart and are detected by surface electrodes.
- ECG machine amplifies the signals and records them as waves, segments and intervals.

Normal conduction pathway:

SA node → AV node → Bundle of His → Right & Left bundle branches → Purkinje fibres

3. ECG Waves



- **P wave** – Atrial depolarization.
- **QRS complex** – Ventricular depolarization.
  - Q wave (first negative)
  - R wave (positive)
  - S wave (negative)
- **T wave** – Ventricular repolarization.
- **U wave** – Sometimes visible (hypokalaemia).

4. Standard 12 Leads

Total 12 leads using 10 electrodes.

Limb leads (6)

Bipolar:
- Lead I
- Lead II
- Lead III

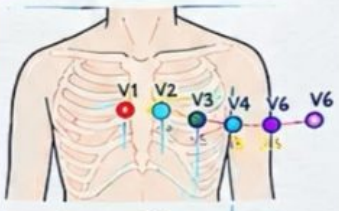
Augmented:
- aVR
- aVL
- aVF

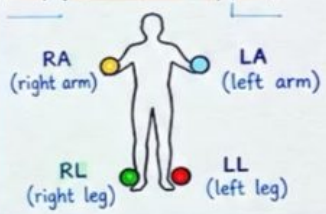
Chest leads (6)
- V1
- V2
- V3
- V4
- V5
- V6

V1–V6 = Chest leads

I, II, III, aVR, aVL, aVF = Limb leads

5. Lead Placement





Chest leads (V1–V6)

- V1 – 4th ICS, right sternal border
- V2 – 4th ICS, left sternal border
- V3 – midway between V2 & V4
- V4 – 5th ICS, midclavicular line
- V5 – same level as V4, anterior axillary line
- V6 – same level as V4, midaxillary line

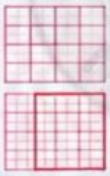
Limb electrodes

- RA – Right arm
- LA – Left arm
- RL – Right leg (ground)
- LL – Left leg

6. ECG Intervals & Segments

Interval / Segment	What it represents	Normal value
PR interval	Atrial → ventricular conduction	0.12 – 0.20 sec
QRS duration	Ventricular depolarization	< 0.12 sec
ST segment	Depolarization → repolarization (early ventricular phase)	Isoelectric (should be flat)
QT interval	Total ventricular depolarization & repolarization	Varies with HR (use QTc)

7. ECG Paper & Calibration



- Small square = 1 mm/nee
  - 0.04 sec (horizontal)
  - 0.1 mV (vertical)
- Large square = 5 mm
  - 0.20 sec (horizontal)
  - 0.5 mV (vertical)

Standard settings
- Paper speed = 25 mm/sec
- Calibration = 10 mm = 1 mV

Heart rate calculation

$$HR = \frac{300}{\text{No. of large squares between two R waves}}$$

Eg: 4 large squares → 300/4 = 75/min

8. Normal Heart Rate & Rhythm

- Normal resting HR (adult) : 60 – 100/min
- Tachycardia : > 100/min
- Bradycardia : < 60/min
- Sinus rhythm : P before each QRS, regular rhythm, normal QRS, constant PR.

9. Common ECG Abnormalities

1. Sinus tachycardia – HR >100/min (exercise, fever, pain, anxiety)
2. Sinus bradycardia – HR <60/min (sleep, athletes, vagal tone)
3. Atrial fibrillation – irregularly irregular, no P waves
4. ST elevation – MI / acute injury
5. ST depression – ischemia, strain
6. T wave inversion – ischemia, strain, electrolyte imbalance
7. Bundle branch block – widened QRS, delayed conduction

12. Importance Physcordial Injctherapy

- Know baseline cardiac status
- Understand heart rhythm
- Monitor exercise response
- Detect arrhythmias
- Ensure safety during rehab

During cardiac rehab (selected cases):

→ ECG/telemetry to monitor HR, rhythm, ischemia.

→ Modify/stop exercise if abnormal response or symptoms occur.

11. Clinical Uses

- Detect arrhythmias
- Asses co HR & & chythm
- Conduify ischemia/infarction
- Conduccice Oguinen
- Chamber safety during rechab

★ Quick Revision

ECG = electrical activity of heart	P = Atria QRS = Ventricles T = Ventricles recover	PR = Atrial to ventricular conduction	12 leads = 6 limb + 6 chest 10 electrodes	Paper speed = 25 mm/sec Calibration = 10 mm = 1 mV Normal HR = 60–100/min

P = Atria, QRS = Ventricles, T = Ventricles Recover

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THANK YOU FOR YOUR PURCHASE!

We wish you the very best in your medical and physiotherapy examinations.

For any questions, supplementary questions, or study support, reach out to us:

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